Krishna Dheeraj Krovi

AI & Backend Software Engineer

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Agentic AI | LangGraph | Model Context Protocol | Retrieval-Augmented Generation | Python | AWS & Azure

SUMMARY

AI & Backend Software Engineer with 4 years of experience building Python services and production LLM systems. Develops multi-agent workflows, secure tool integrations, hybrid retrieval, and deterministic evaluation, with AWS and Azure deployment experience. Builds execution controls, streaming interfaces, and observability for reliable AI operations.

SKILLS

Languages: Python, Go, SQL, Java, JavaScript, Bash
Agentic AI: LangGraph, LangChain, Model Context Protocol (MCP), Tool Calling, Human-in-the-Loop
Retrieval & Models: RAG, Embeddings, pgvector, Milvus, Knowledge Graphs, OpenAI, Anthropic, Gemini, Ollama
AI Reliability: Checkpoint/Resume, Bounded Retries, Token Budgets, Structured Outputs, Offline Evaluation, Grounding
Backend & Data: FastAPI, REST APIs, Server-Sent Events (SSE), React, Next.js, PostgreSQL, Redis, Elasticsearch
Cloud & Delivery: AWS, Azure Functions, Service Bus, Blob Storage, Bicep, Docker, Kubernetes, Azure DevOps
Testing & Observability: pytest, Ruff, Bandit, Gitleaks, OpenTelemetry, Prometheus, CloudWatch, Grafana

WORK EXPERIENCE

Artha Solutions · Software Engineer *May 2022 - Present*

- Architected a multi-agent LLM conversion pipeline with analyzer, planner, generator, reviewer, and patcher stages; gated generated code with deterministic AST, lint, and rule-based checks.
- Built LangGraph workflows with PostgreSQL checkpoints, human approval and escalation interrupts, bounded retries, timeouts, and token and cost budgets for resumable agent execution.
- Implemented a Python MCP server with OAuth 2.1 and PKCE, per-user scopes, and versioned tool allowlists to expose knowledge and database tools to agent clients.
- Built OpenAI-compatible and Ollama tool-calling runners with guarded file, command, search, memory, and database operations; added repeat-call refusal, turn limits, and model routing.
- Created hybrid RAG with full-text search, pgvector, reciprocal-rank fusion, context-aware chunking, and a knowledge graph; achieved 0.94 recall@5 and 0.87 MRR in offline retrieval evaluation.
- Designed LLM evaluation and grounding checks with retrieval ledgers, citation gates, and threshold-based datasets; automated 44 guard cases and five end-to-end scenarios in CI.
- Built React/Next.js agent interfaces and FastAPI SSE APIs for streaming output, tool activity, approvals, steering, cancellation, checkpoint resume, and token and cost visibility.
- Orchestrated AI execution with AWS Step Functions and Bedrock AgentCore, using checkpointed steps, stall detection, human escalation, bounded retries, and per-job token ceilings.
- Instrumented agent runs with OpenTelemetry traces, structured logs, Prometheus metrics, and Grafana dashboards; deployed services using AWS, Docker, Kubernetes, and CI/CD safety gates.
- Migrated Boomi integrations to Python Azure Functions, preserving API routes and queue-triggered workflows; integrated Service Bus, Blob Storage, and Key Vault with managed identities and scoped RBAC.
- Built GitHub Actions/JFrog build workflows and Azure DevOps release automation with pinned internal wheels, artifact checks, API-key configuration, and idempotent Service Bus role assignments.

Wipro Limited · Project Engineer *Dec 2020 - Jun 2021*

- Developed and tested Java REST API integrations using Postman and MVC architecture, improving service validation and reducing manual effort.
- Built SQL data-extraction workflows for reporting and operational analysis; supported production issue resolution, releases, and user acceptance testing.

EDUCATION

M.S., Electrical Engineering, University of Southern California, Los Angeles, USA
M.Tech & B.Tech, Electronics & Communications Engineering, JNTU, Hyderabad, India

CERTIFICATIONS

- AWS Certified Machine Learning - Specialty, Amazon Web Services
- Claude Certified Architect - Foundations, Anthropic